AXEL KOEHLER · DISTINGUISHED ENGINEER, NVIDIA · GERMANY

From a Physics Lab to Exascale: 34 Years in HPC



Four employers, one constant: helping scientists get the most out of the biggest machines

1992 — DESY Zeuthen
System Engineer
4 yrs 8 mos
Running computing for a particle-physics research lab

1996 — Sun Microsystems
Senior HPC Architect
13 yrs 9 mos
Designing HPC systems for research & industry

2010 — Oracle
Principal Sales Consultant
6 mos
Transition via the Sun acquisition

2011 — NVIDIA
Principal Solution Architect
14 yrs 3 mos
Bet on GPU computing – from early GPUs to AI factories

2025 — NVIDIA
Distinguished Engineer
since Mar 2025
Technical leadership for Europe's flagship centres

Rooted in academia: Technical University of Dresden · **Eras lived through:** SMP servers → commodity clusters → GPU acceleration → exascale & AI factories

TECHNICAL JOURNEY & CAREER TRANSITIONS

How the Expertise Grew – and Where Careers Turned

1 Learn the system end-to-end

Research lab · 1992–1996

- Operating computing at a physics lab taught me how scientists actually use machines
- Hands-on: OS, networks, storage, user support
- Lesson: understand the workload before the hardware

2 Architect at scale

Sun Microsystems · 1996–2010

- Grew from systems to full HPC architectures for universities, labs & industry
- Benchmarks, procurements, system design
- Turn: the Oracle acquisition (2010) forced a rethink of where HPC was heading

3 Bet on acceleration

NVIDIA · 2011–today

- Joined when GPU computing was still a niche bet
- Partner to centres such as JSC, LRZ, HLRS, CSCS, MPCDF and to industry (e.g. Airbus)
- From porting & benchmarking to co-designing exascale systems and AI factories

Leadership without a big title: influence through trust, technical depth and long-term relationships with customers, partners and peers

A EUROPEAN PERSPECTIVE

Five Lessons for Rising Innovators

1. **Bet on the next architecture early**
   Moving to GPUs in 2011 looked risky. Disruptive shifts reward those who learn them first.

2. **Stay close to the applications**
   Benchmarks and real codes beat datasheets. Credibility comes from solving users' problems.

3. **Treat change as a constant**
   Acquisitions, re-orgs, new architectures: fundamentals transfer – keep learning.

4. **Energy efficiency is the new performance**
   Sustainable exascale & AI means performance per watt, not just peak flops.

5. **Build bridges across sectors**
   Academia, national labs, industry and vendors – the best systems are co-designed together.

Let's talk!

Happy to chat about:

- HPC & AI system architecture
- Large-scale GPU clusters
- Moving from academia to industry
- Technical career paths

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“The machines change every few years – curiosity, fundamentals and relationships last a career.”